RentAFit Virtual Try-On Setup Guide
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CatVTON on a Blank Windows NVIDIA PC

Prepared for: RentAFit project
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1. Purpose of This Guide

This guide explains, step by step, how to run **CatVTON** on a Windows PC that currently has **nothing installed for coding**.

This is written for a beginner. It assumes: - the PC is running Windows 10 or Windows 11 - the PC has an NVIDIA GPU - the user is not comfortable with Windows terminal setup yet - the goal is to **run virtual try-on locally**, not train a new model

This guide focuses on **inference only**.

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2. What CatVTON Will Do for the Project

CatVTON is the virtual try-on model chosen for the RentAFit project.

It takes: - a **person image** - a **garment image**

and generates: - an output image showing the person wearing the garment

For the current RentAFit project scope, the best starting categories are: - Shirt - Top - Jacket

These categories are simpler and more reliable than saree, lehenga, or more heavily draped clothing.

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3. Why This Guide Uses the Official CatVTON App Route

This guide uses the **official CatVTON repository** and its **local Gradio app**.

This is the best first attempt because: - the official repository documents this workflow directly - the official app can auto-download checkpoints on first run - the project notes that DensePose and SCHP were localized to reduce environment issues - this is cleaner for a beginner than starting with custom ComfyUI node debugging

This is an informed recommendation based on the official project documentation.

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4. Very Important Warning Before You Start

If the Windows PC has one of these GPUs: - RTX 5080 - RTX 5090

stop before installing and re-check the environment setup, because very new GPUs sometimes need slightly different PyTorch or CUDA handling.

If the GPU is one of these, this guide is usually appropriate: - RTX 3060 / 3070 / 3080 / 3090 - RTX 4060 / 4070 / 4080 / 4090

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5. What You Need Before Installation

Before you start, make sure you have: - a stable internet connection - at least 30 GB of free disk space - an NVIDIA GPU with enough VRAM - clean test images ready - enough time for downloads, because the first run can take quite a while

The official CatVTON project notes that `1024 x 768` inference with `bf16` can run in under `8 GB` VRAM.

In practice, more VRAM is still better.

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6. Prepare the Test Images First

Before installing anything, prepare a small clean test set.

6.1 Person images

Prepare at least 5 person photos.

Each should be: - front-facing - upper body clearly visible - well lit - not blurry - with a simple or clean background - with the torso not blocked heavily by arms, bags, or scarves

6.2 Garment images

Prepare at least 5 garment photos.

Each should be: - a single garment only - front-facing if possible - clearly visible - not folded badly - well lit - not a collage - not a screenshot with text over it

6.3 Categories to start with

Use only: - Shirt - Top - Jacket

Do not start with: - Saree - Lehenga - heavily layered ethnic sets

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7. Create Folders on the Windows PC

Create a clean folder structure first.

Example root folder:

```
D:\RentAFit_TryOn
```

Inside that folder, create:

```
D:\RentAFit_TryOn\inputs\person
D:\RentAFit_TryOn\inputs\garment
D:\RentAFit_TryOn\outputs
D:\RentAFit_TryOn\notes
D:\RentAFit_TryOn\software
```

Put your test images here:

```
D:\RentAFit_TryOn\inputs\person\person_01.jpg
D:\RentAFit_TryOn\inputs\person\person_02.jpg
D:\RentAFit_TryOn\inputs\garment\shirt_01.jpg
D:\RentAFit_TryOn\inputs\garment\top_01.jpg
D:\RentAFit_TryOn\inputs\garment\jacket_01.jpg
```

Use simple names. Avoid spaces and strange symbols.

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8. Check the GPU Model on Windows

Do this before anything else.

Method

1. Press `Ctrl + Shift + Esc`
2. Open **Task Manager**
3. Click **Performance**
4. Click **GPU**
5. Note the exact GPU model name

Write this down in your notes.

Example: - NVIDIA GeForce RTX 4070

If the GPU is 30-series or 40-series, this guide is usually fine.

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9. Update the NVIDIA Driver

This step is very important.

What to do

1. Open the official NVIDIA driver page: NVIDIA Driver Downloads
2. Select the exact GPU model
3. Download the latest driver
4. Install it
5. Restart the computer

Do not skip the restart.

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10. Install Miniconda

The official CatVTON setup uses a conda environment, so this guide uses **Miniconda**.

Official links

- Miniconda install guide
- Miniconda download page

What to do

1. Open the Miniconda download page
2. Download the **Windows 64-bit installer**
3. Run the installer
4. Choose **Just Me** unless you specifically need all users
5. Finish installation
6. Open **Anaconda Prompt** from the Start menu

Use **Anaconda Prompt** for the rest of the setup.

Do not worry about regular Command Prompt for now.

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11. Download the Official CatVTON Repository

Official source

- CatVTON official GitHub repository

What to do

1. Open the GitHub page
2. Click the green **Code** button
3. Click **Download ZIP**
4. Save the ZIP file into:

```
D:\RentAFit_TryOn\software
```

5. Extract it
6. Rename the extracted folder to:

```
D:\RentAFit_TryOn\software\CatVTON
```

Avoid spaces in the folder path.

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12. Open Anaconda Prompt and Create the Environment

Open **Anaconda Prompt**.

Then run this command:

```
conda create -n catvton python=3.9.0 -y
```

Wait for it to finish.

Then activate the environment:

```
conda activate catvton
```

You should now see `(catvton)` at the beginning of the command line.

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13. Move Into the CatVTON Folder

Still inside **Anaconda Prompt**, go to the extracted CatVTON folder:

```
cd /d D:\RentAFit_TryOn\software\CatVTON
```

The `/d` is important on Windows because it changes the drive as well.

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14. Install GPU PyTorch First

The official CatVTON requirements pin these versions: - `torch==2.1.2` - `torchvision==0.16.2`

To make sure the install uses the NVIDIA GPU properly, install PyTorch first using the official PyTorch command.

Official source

- PyTorch previous versions page

Command

```
conda install pytorch==2.1.2 torchvision==0.16.2 torchaudio==2.1.2
        pytorch-cuda=12.1 -c pytorch -c nvidia -y
```

Wait for this to finish fully.

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15. Install the Remaining CatVTON Requirements

Now install the rest of the project dependencies.

Inside the same Anaconda Prompt window, still in the CatVTON folder, run:

```
pip install -r requirements.txt
```

Official requirements reference

- CatVTON requirements.txt

This step can also take some time.

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16. Verify That CUDA Is Available

Before running the app, check whether Python can actually see the GPU.

Run this command:

```
python -c "import torch; print('CUDA available:',
        torch.cuda.is_available()); print('GPU count:',
        torch.cuda.device_count()); print('GPU name:',
        torch.cuda.get_device_name(0) if torch.cuda.is_available()
        else 'None')"
```

Expected result

You want to see something like: - `CUDA available: True` - `GPU count: 1` - your GPU name printed correctly

If you get `False`, do not continue yet.

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17. Start the Official CatVTON App

The official CatVTON README provides this command for the local app:

```
python app.py --output_dir="resource/demo/output" --
        mixed_precision="bf16" --allow_tf32
```

Important first-run note

The official repository says the app can auto-download the checkpoints the first time it runs.

That means: - first startup can be slow - downloads can be large - it can take many minutes

Do not assume it is broken too early.

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18. If `bf16` Fails, Use `fp16`

If the app gives a precision-related error, close it and run this instead:

```
python app.py --output_dir="resource/demo/output" --
        mixed_precision="fp16" --allow_tf32
```

This is the safer fallback.

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19. Open the Local Web App

Once the app starts successfully, the terminal should print a local web address.

Usually it will look something like: - `http://127.0.0.1:7860`

Open the exact address printed in the terminal in the browser on that same PC.

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20. Run Your First Test Only

Do not try many images first.

Use only: - 1 person image - 1 garment image

Example first test: - `person_01.jpg` - `shirt_01.jpg`

Inside the app

1. upload the person image
2. upload the garment image
3. leave most settings at default for the first run
4. generate the try-on result

The goal of this first test is simple: - prove the pipeline runs successfully

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21. Save the Output Properly

When the first run completes:

1. save the generated output image
2. copy or move it into:

```
D:\RentAFit_TryOn\outputs
```

Use a clean filename such as:

```
test_01_person_01_shirt_01.png
```

Also create a text or spreadsheet note in:

```
D:\RentAFit_TryOn\notes
```

Suggested columns: - `test_id` - `person_image` - `garment_image` - `category` - `status` - `quality_note`

Example:

```
T001, person_01.jpg, shirt_01.jpg, Shirt, success, Good overall fit,
sleeve edges slightly soft
```

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22. Run 5 to 10 Small Tests After the First Success

Once the first run works, test a small batch.

Recommended order: - 2 Shirt cases - 2 Top cases - 1 Jacket case

Then expand if results look good.

This is enough for: - project documentation - demonstration - comparison of good and bad results

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23. Troubleshooting: CUDA Is Not Available

If the verification command says: - `CUDA available: False`

then check these in order:

1. Did you update the NVIDIA driver?
2. Did you restart the PC?
3. Did you open **Anaconda Prompt**, not normal terminal?
4. Does the prompt show `(catvton)`?
5. Did the PyTorch install command complete successfully?

If this still fails, note the exact error and stop there.

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24. Troubleshooting: Out of Memory

If you get an out-of-memory error:

1. close Chrome tabs and other heavy apps
2. rerun using `fp16` instead of `bf16`
3. if the app offers output size settings, reduce the size for testing
4. do not run multiple GPU-heavy tools at the same time

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25. Troubleshooting: First Run Takes Too Long

This is often normal.

Possible reasons: - checkpoint download - slow internet - initial model loading - large dependency initialization

If the terminal is still active and not crashing, wait longer before interrupting it.

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26. Troubleshooting: Output Quality Is Poor

Poor output quality is often caused by poor inputs, not broken installation.

Check whether: - the person image is too dark - the garment image is cluttered - the garment is not clearly visible - the pose is awkward - the torso is partly blocked - the photo is blurry

Before changing code, improve the images.

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27. What You Should Not Do Right Now

Do not: - train CatVTON from scratch - start with saree or lehenga - jump into website integration before the local demo works - test too many categories at once - use random bad-quality images first

Keep the first milestone small and controlled.

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28. Recommended Documentation You Should Keep

For your college project, save these: - GPU model used - RAM size if known - number of test cases - successful outputs - failed outputs - observed strengths - observed failure cases

This will help later in the final report and viva.

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29. Best Workflow Summary

Use this exact order:

1. check GPU model
2. update NVIDIA driver
3. create folders
4. prepare 5 person images and 5 garment images
5. install Miniconda
6. download CatVTON ZIP
7. create the `catvton` environment
8. install PyTorch with CUDA support
9. install CatVTON requirements
10. verify CUDA is available
11. run the app
12. do 1 test
13. save the result
14. do 5 to 10 tests
15. document everything

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30. Fallback Option If This Fails on Windows

If the official `app.py` route fails badly on Windows, the fallback is: - ComfyUI + CatVTON

Official references for that route: - ComfyUI official docs - Official CatVTON Windows note issue #8

Use this fallback only if the direct app route becomes unmanageable.

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31. License Note

The CatVTON model page lists the materials under: - `CC BY-NC-SA 4.0`

That means it is suitable for a non-commercial academic project, but not something to assume is unrestricted for commercial use.

Official source: - CatVTON Hugging Face page

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32. Official Sources Used

- CatVTON official GitHub
- CatVTON official Hugging Face page
- Miniconda official install guide
- PyTorch previous versions page
- CatVTON Windows note issue #8
- ComfyUI official docs

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33. Final Note

The correct goal for the first session on your friend's PC is not perfection.

The correct goal is: - get one clean successful try-on run - save it properly - then expand carefully

Once that first success is done, the rest becomes much easier.